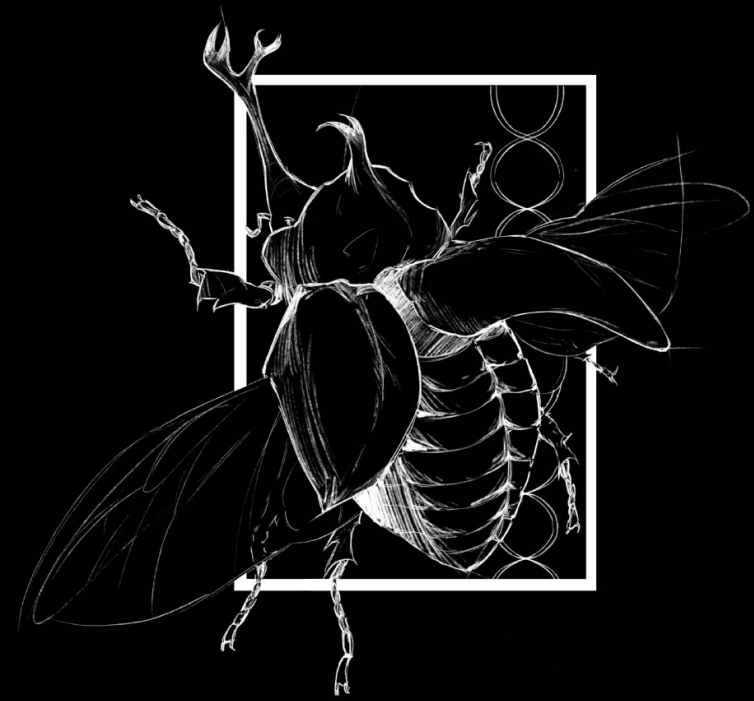




Introduction

Spider silk has many possible industrial and textile applications. Understanding the factors in play regarding silk strength is critical when investing effort into silk production for human applications. Determining the strength of dragline silk in a wide range of spider species and evaluating the data through an evolutionary lens provides a valuable insight into the possible factors that affect tensile strength in spiders.

Previous studies determined that tensile strength across the family Araneae did not correlate with silk usage in webs and hunting behavior. **This study aimed to determine whether spider silk evolution across the Araneae clade moves towards an optimal strength rather than one determined by environmental factors or silk usage.**

Study Design & AI

- As part of the Blackmon Lab, the use of AI was encouraged to assist in code creation and data analysis. Claude Opus 4.5 was utilized through Texas A&M University’s TAMU AI Chat function with guidance from Dr. Heath Blackmon.
- Tensile strength data for 277 species from the clade Araneae was obtained from the *Spider Silkome Database* at <https://spider-silkome.org> in 2026. The *Spider Silkome Database* serves as a repository for data regarding spider dragline silk physical properties, silk morphologic structure, and spidroin sequencing.
- The phylogeny of Araneae was obtained from research conducted by Jonas Wolff regarding the evolution of spider silk with respect to its usage across Araneae. Wolff determined that silk strength has not evolved as a result of its usage across species in different web structures.
- With the help of Claude, tensile strength data from the *Spider Silkome Database* was plotted against the phylogeny of the clade Araneae (Figure 1).

Model Fitting

Data used in this study was analyzed under both a Brownian and Orstein-Uhlenbeck model to determine if spider silk is subject to balancing selection.

Previous studies utilize Brownian motion to determine whether spider silk evolves towards specific use cases by treating each evolutionary movement as random. Orstein-Uhlenbeck modelling evaluates the movement of that trait to detect the presence of a possible optimal value over time.



Results

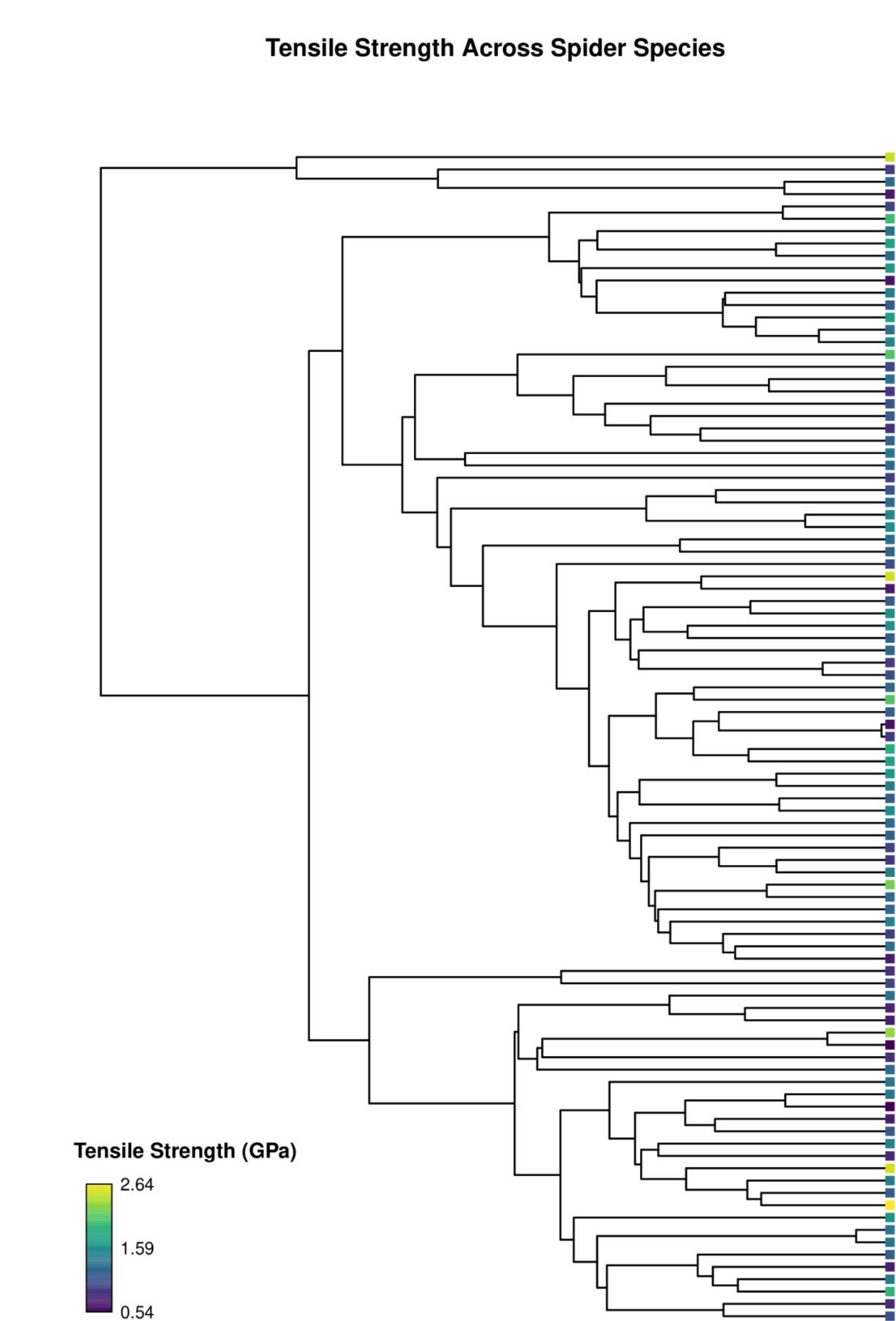


Figure 1 Tensile Strength of Silk. The squares at the tip of the phylogeny illustrate the tensile strength of drag line silk in each species ranging from just 0.54 GPa to 2.64 GPa

When analyzed, the Ornstein-Uhlenbeck process yielded an alpha value of 0.06804, indicating strong purifying selection. When compared, the difference in AIC between Brownian Motion and Ornstein-Uhlenbeck modelling is 47.6258, suggesting that the evolution of spider silk strongly follows an Ornstein-Uhlenbeck model. The figure (left) displays the distribution of tensile strength across the clade Araneae.

The table below shows comparative measurements of the tensile strength of *Triconephila inaurata* and *Argiope trifasciata* between their dragline silk and minor ampullate silk.

Species	Major Ampullate (Dragline) Silk Strength (GPa)	Minor Ampullate Silk Strength (GPa)
<i>Triconephila inaurata</i>	1.73	1.5
<i>Argiope trifasciata</i>	1.3	1.04

As seen in the data and known generally, minor ampullate silk tends to be thinner and weaker than dragline silk, but a lack of data prevents similar modelling to see if trends follow dragline silk or if there may be more variation with non-dragline silk types.



Discussion

Spider silk exhibits signs of balancing evolution towards a optimal tensile strength. This shows evidence of a possible evolutionary pressure to keeping spider silk at that value.

Analyses of non-dragline silks such as minor ampullate silk, flagelliform silk, and tubuliform silk remains scarce and lacks the same depth of research given to dragline silk. Without effective research towards the strength of these silk types, it remains unclear whether the purifying selection of silk tensile strength is a result of environmental pressure regarding web structure or atmospheric factors or if purifying selection is a result of the physiological limitations of spinnerettes.

AIC values represent the Akaike Information Criterion, which evaluates the likelihood of the model presented against the complexity required to use that model. When evaluating AIC values, lower values are preferred over higher ones, as this indicates models that fit better or are less complex.

Brownian Motion evaluates the evolution of a trait as it moves randomly through time. Previous studies have mapped dragline silk strength against a Brownian Motion model to see if the trait experienced evolution with respect to the silk’s use case. While this model is helpful, it does not consider whether silk strength is selected towards an ideal value.

Ornstein-Uhlenbeck modelling examines this particular niche, evaluating whether silk strength evolves towards an optimal value instead of random movement. This form of analysis can reveal possible limitations in trait evolution or selective pressures that exist across an entire clade.

The Ornstein-Uhlenbeck AIC value was significantly smaller than the Brownian Motion AIC value. This large difference suggests that Ornstein-Uhlenbeck modelling is strongly preferable to Brownian Motion. Applying these findings to our data reveals that spider silk must evolve towards an optimal tensile strength rather than radiating outwards randomly.

Conclusions & Acknowledgements

Spider silk evolves towards a optimal strength through balancing selection. It is not yet clear what selective pressures exist to cause these results.